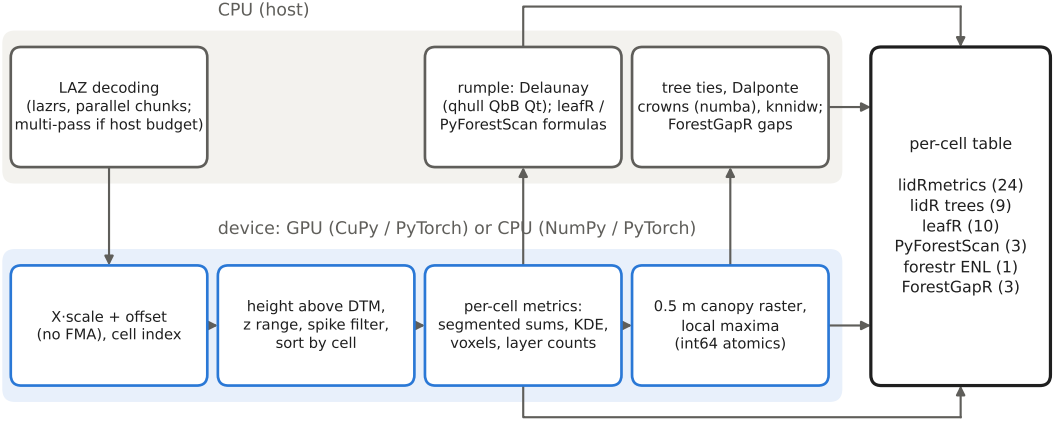with a memory budget the device stages run on blocks of cell rows